

Geography Ch-19 (Must-Do): The Warm Temperate Western Margin (Mediterranean) Climate (Cs)

Tier 1: Must-Do | Source: GC Leong + Majid Husain (Indian & World Geography)
Grounded in Qdrant upsc_static book DB + live Current Affairs anchor

Facts from official sources | Inferences clearly marked | No fabricated data

HIGH

The Mediterranean Climate (Cs): Dry Sunny Summers, Wet Mild Winters & the Fruit Economy

GS-I (Physical
Geography -
Climatology)
Geography

■ NEWS TRIGGER

Foundation concept (GC Leong Ch-19). The Mediterranean climate occupies the western margins of continents in the warm-temperate zone (30-45 deg N & S) - typified by dry, sunny summers and wet, mild winters. Koppen Cs.

■ INTRO & ORIGIN

The Mediterranean is the 'winter-rain' climate: in summer the sub-tropical high shifts poleward over it (dry, sunny, hot); in winter the westerlies and their depressions shift equatorward onto it (mild, rainy). This unique summer-drought/winter-rain reversal shapes its scrub vegetation and world-famous fruit-and-wine agriculture.

Caused by the SEASONAL SHIFT of pressure/wind belts: in summer the region lies under the Sub-tropical High (offshore trades, subsidence, drought); in winter it lies within the belt of Westerlies and travelling depressions (onshore rain).

■ KEY DATA

Feature	Detail	Example
Location	West margins, 30-45 deg N & S	Mediterranean basin, California, C. Chile, Cape, SW Australia
Summer	Hot, dry, sunny (Sub-tropical High)	Drought; clear skies
Winter	Mild, wet (Westerlies + depressions)	Most rain in winter
Vegetation	Drought-resistant scrub & woodland	Maquis, chaparral; olive, cork oak
Agriculture	Orchard/fruit + wine (Mediterranean)	Citrus, olives, grapes; winter wheat

■ STATIC THEORY

- **The defining reversal:** DRY summers + WET winters - the only climate with a pronounced summer drought in the temperate zone; caused by the poleward (summer) and equatorward (winter) migration of the pressure/wind belts.
- **Local winds (GC Leong):** hot, dusty winds from the deserts (Sirocco of the Sahara, also Khamsin) and cold, stormy winds from the interior/mountains (Mistral down the Rhone valley, Bora on the Adriatic).
- **Vegetation - sclerophyllous scrub:** drought-adapted, small broad-leaved, widely spaced trees; NO shade (a distinctive feature). Local scrub names: maquis (Mediterranean), chaparral (California), mallee (Australia). Species: olive, cork oak, fig, eucalyptus; deep roots, waxy/leathery leaves, thick bark.
- **Growth timing:** because summer is dry and winter cool, plant growth is concentrated in the AUTUMN and SPRING (warm + moist enough).
- **Agriculture (Majid Husain):** the Mediterranean agricultural system specialises in ORCHARDS of citrus fruits (cereals subordinate); winter wheat and barley; and viticulture - the region produces about three-quarters of the world's wine.

■ MUST-KNOW FACTS

1. Koppen Cs; western margins, 30-45 deg; dry sunny SUMMER + wet mild WINTER.
2. Caused by seasonal migration of pressure belts (Sub-tropical High in summer, Westerlies in winter).
3. 5 regions: Mediterranean basin, California, central Chile, SW Cape (S. Africa), SW & S Australia.

■ UPSC TRAPS

- ✗ The Mediterranean climate has summer rainfall like the monsoon.
- ✓ It has SUMMER DROUGHT and WINTER RAIN - the reverse of the monsoon; rain comes from winter westerly depressions.
- ✗ Sirocco and Mistral are both hot winds.

4. Scrub names: maquis (Med), chaparral (California), mallee (Australia); olive & cork oak typical.

5. World's leading fruit-and-wine region (~3/4 of global wine); Sirocco (hot) & Mistral (cold) winds.

✓ Sirocco is a HOT, dusty wind from the Sahara; Mistral is a COLD, stormy wind blowing down the Rhone valley.

✗ Mediterranean vegetation is dense shady forest.

✓ It is widely-spaced, small-leaved drought-resistant scrub/woodland; the ABSENCE of shade is a hallmark feature.

■ MAINS ANGLE

Explain how the seasonal migration of pressure belts produces the Mediterranean summer-drought/winter-rain regime, and how this shapes its scrub vegetation and specialised fruit-wine agriculture.

■ STUDY LINK

Geography -> World Climatic Types -> Mediterranean (Cs); links to Westerlies, sub-tropical high & horticulture

■ MEDIUM UPSC RELEVANCE

MEDIUM

CA Anchor: 2025 Mediterranean Wildfires - Climate Change Supercharges Summer Drought

GS-I (World Geography) +
GS-III (Disaster Management / Climate Change)
Geography -
Current Affairs

■ NEWS TRIGGER

2025 was Europe's worst wildfire season on record - over 1 million hectares burned across Greece, Spain, Italy, Turkey, Cyprus and southern France amid 40-45 deg C heat and drought. Rapid-attribution science found the hot, dry, windy conditions were made ~10x more likely by climate change; fires released ~38 million tonnes of CO₂.

■ INTRO & ORIGIN

The Mediterranean's built-in SUMMER DROUGHT makes it the world's most fire-prone temperate climate - and 2025 shows how climate change is turning that natural dryness into catastrophic, self-reinforcing wildfire seasons.

The chapter's 'dry summer + flammable sclerophyll scrub' is now amplified by longer, hotter, drier summers. Attribution studies quantify the climate-change fingerprint.

■ KEY DATA

Metric	2025 value	Note
Area burned (Europe)	> 1 million hectares	Worst on record
Climate attribution	~10x more likely	Hot-dry-windy conditions
Heat	40-45 deg C	Prolonged Mediterranean heatwave
Emissions	~38 million tonnes CO ₂	Feedback loop

■ STATIC THEORY

- **Why the Mediterranean burns:** summer drought dries the sclerophyllous scrub (maquis/chaparral) - naturally flammable, oil-rich leaves - and hot, dry local winds (Sirocco-type) fan fast-spreading fires.
- **Climate-change amplification:** longer/hotter/drier summers extend the fire season and raise intensity; rapid-attribution science now estimates how much more likely such conditions are.
- **Fire-emission feedback:** wildfires release large CO₂ pulses (~38 Mt in 2025), which further warms the climate - a vicious loop.

■ MUST-KNOW FACTS

1. 2025 = Europe's worst wildfire season on record; >1 million ha burned across the Mediterranean.
2. Attribution: hot-dry-windy fire weather ~10x more likely due to climate change.
3. Heatwave temperatures reached 40-45 deg C; ~38 Mt CO₂ released by the fires.
4. Mediterranean summer DROUGHT + flammable maquis/chaparral scrub = high fire risk.
5. Longer fire seasons projected; Mediterranean wildfires may rise ~50% by 2100 on current trends.

■ UPSC TRAPS

- ✗ Mediterranean wildfires peak in winter (the rainy season).
- ✓ They peak in SUMMER, the dry drought season, when scrub is desiccated and winds are hot and dry.
- ✗ Wildfires have no feedback on climate.
- ✓ They emit large CO₂ pulses (~38 Mt in 2025), reinforcing warming - a positive feedback loop.

■ MAINS ANGLE

Use the 2025 Mediterranean wildfires to explain how climate change amplifies the natural summer-drought fire risk of the Mediterranean climate, and the disaster-management response needed.

■ STUDY LINK

Geography -> Mediterranean climate (summer drought) + GS-III DM/climate change (wildfires, attribution science)

End of Register Notes — UPSC Agent / Copilot CLI